Chapter 4

Compositional and Chemical Control of Volatile Polymer-Electrolyte Memristive Dynamics

4.1 Introduction and Scope

Chapter 2 established the memristive response of the formulation comprising *Super Yellow*, *Hybrane*, and LiOTf and of its 2-T architecture across assay-specific junctions. The broader device corpus examined here varies the composition, electrolyte chemistry, electrode, and drive protocol of the same three-component architecture.

Three families of measurement recur on the same 2-T geometry throughout this chapter and the next: (i) current–voltage hysteresis, which reports the steady-state switching window; (ii) potentiation as a function of the number of applied write pulses, which reports how the conductance is built up; and (iii) depotentiation as a function of the delay before read-out,

which reports how a written state fades. Introduced as a named set in chapter 2, they are the shared, abundant basis of the comparative work. The richer proof-of-concept synaptic protocols of chapter 2 (excitatory post-synaptic current, separated short- and long-term memory, and spike-timing dependent plasticity) are *not* repeated here; where invoked, they serve only as formulation-level prior evidence.

Scope of the comparisons. The *Super Yellow*/PEO/LiOTf pulse and decay grids contain two to four nominally identical substrates per cell; the steady-state HYST corpus contains additional compatible batches and therefore two to thirteen substrates per cell. These replicated datasets support quantitative composition comparisons. After data-quality screening, the host, anion, and cation contrasts contain at most one or two substrates per matched cell and are used only as an illustrative survey. The electrode comparison is likewise illustrative and is available only at the lead composition (section 4.8). Drive protocol is analysed separately because it changes the apparent fading-memory timescale and therefore limits comparisons between protocols.

4.2 Materials, Devices, and Measurement Protocol

The devices retain the architecture of chapter 2: a vertical two-terminal sandwich of ITO (bottom, transparent), a solution-processed composite active layer, and a thermally evaporated silver top electrode, with a junction area of $8.25 \times 10^{-2}\,\mathrm{cm}^2$. *Super Yellow* is retained as the semiconducting component in every device of this chapter, so that the electronic-transport role is held fixed and the variation is confined to the ion-transport phase.

The experiment varies composition and chemistry. The *composition* axis

changes the mass ratios of the ion-transport polymer poly(ethylene oxide) (PEO) and lithium triflate salt (LiOTf) relative to the semiconductor: $r_{PEO} = m_{PEO}/m_{Super\ Yellow}$ and $r_{\mathrm{salt}} = m_{\mathrm{salt}}/m_{Super\ Yellow}$. These are blend-component ratios, not fractions of the total dry mass. The *chemistry* axis substitutes the ion-transport host, the salt anion (triflate or bis(trifluoromethanesulfonyl)imide, TFSI), and the alkali cation (Li^+, Na^+, K^+, as the triflate or TFSI salt). The comparative work replaces the hyperbranched polyester-amide *Hybrane* host of the proof-of-concept device (chapter 2) with two hosts: linear PEO and the hyperbranched polyether trimethylolpropane ethoxylate (TMPE). This gives a matched-chemistry contrast between linear and branched polyethers. Every device reported *quantitatively* uses the silver electrode. A separate gold-electrode corpus at the lead composition compares an active with an inert electrode, measures drive-polarity dependence on gold, and provides an independent-electrode check on the chemistry comparisons (section 4.8).

Film thickness as a measured covariate. Adding ion-transport polymer thickens the solution-processed film, so most higher-PEO cells were spun faster to reduce thickness differences. Spin speeds across the grid ranged from 1500 rpm to 3500 rpm. This compensation was incomplete and was not applied uniformly to every comparison batch. Among the 28 replicated-grid substrates with both profilometry and a dynamical measurement, median thickness rises from ≈ 227 nm at $r_{PEO} = 0.3$ to ≈ 298 nm at $r_{PEO} = 1.2$ (Pearson $r = +0.65$; fig. 4.1). The activation voltage shows no corresponding thickness trend ($r \approx -0.06$), and a 38% thickness spread within the lead cell leaves $t_{1/2}$ within 18 s–22 s. In the 23 substrates with a model-free $t_{1/2}$, the partial correlation of thickness with $\log t_{1/2}$ at fixed r_{PEO} is $+0.28$, whereas that of r_{PEO} with $\log t_{1/2}$ at fixed thickness remains -0.51. Thickness therefore does not account for the composition association, although the sample size does not exclude a secondary thickness contribution. It is retained as a measured covariate

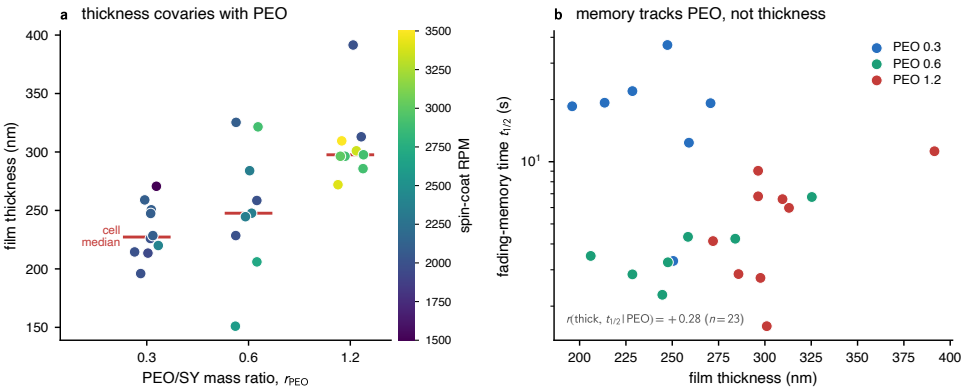


Figure 4.1: Film thickness and fading-memory time in the replicated silver composition grid. (a) Film thickness rises with r_{PEO} (red bars are cell medians; $n = 28$ substrates with profilometry and a dynamical measurement); higher-PEO cells were spun faster (colour) to reduce this difference, although residual covariation remains. (b) For the $n = 23$ substrates with $t_{1/2}$, the partial correlations are $r_{\text{partial}}(\text{thickness}, \log t_{1/2} \mid r_{PEO}) = +0.28$ and $r_{\text{partial}}(r_{PEO}, \log t_{1/2} \mid \text{thickness}) = -0.51$. Thickness does not remove the composition association, but a secondary thickness contribution is not excluded.

rather than declared fully controlled; the complete audit is given in Appendix C.

Drive and read protocols. Because the comparisons below turn on them, the electrical protocols are stated explicitly, and they are *not* identical across the three measurements (table 4.1). Hysteresis uses a triangular voltage sweep. The potentiation (N-pulse build-up) was acquired with a 3 V write and 1.5 V read at a variable pulse count ($N \leq 1000$); the depotentiation (fading-memory decay) used a fixed train of 30 write pulses at a 4 V write and 2 V read. The inter-pulse interval is effectively common (0.104 s for potentiation, 0.103 s for depotentiation), but the *write amplitude differs* between the two (3 V vs 4 V), a point that matters when the two are later composed (section 4.10, Chapter 5). The read voltage is above the ionic activation threshold established in chapter 2,

Table 4.1: The three common measurements use distinct drive protocols (modern *Super Yellow*/PEO/LiOTf/Ag corpus). Potentiation and depotentiation differ in write amplitude, so the chapter never pools them and Chapter 5 composes them only as an explicit assumption.

Measurement	Write	Read	Pulse train
Hysteresis (HYST)	triangular sweep	—	—
Potentiation (PULSES)	3 V	1.5 V	variable $N \leq 1000$, 0.104 s
Depotentiation (DELAYTIME)	4 V	2 V	30 pulses, 0.103 s
Protocol overlay (v114)	3 V, 6 V	1.5 V, 3 V	30 pulses

a point returned to in section 4.6. An older 2022 device generation used a 6 V write with a 3 V read; the modern and old amplitudes are never pooled, and the same-device contrast between two amplitudes is itself the drive-protocol result (section 4.6).

4.3 Methods of Analysis

Hysteresis curves are summarised by the on–off conductance ratio and by a normalised loop area; potentiation curves by the per-decade growth of the conductance ratio with pulse number, by the peak ratio, and by whether the response turns over (decreases) at the highest pulse counts; and depotentiation curves by a fading-memory time constant.

For the depotentiation (fading-memory) decay, the comparative archive is refitted with the more general Kohlrausch (stretched-exponential) function

$$\Delta G(t_{\mathrm{wait}}) = \Delta G_0 \exp\left[-\left(\frac{t_{\mathrm{wait}}}{\tau}\right)^{\beta}\right] + \Delta G_{\infty}, \tag{4.1}$$

with characteristic relaxation time τ, stretch exponent $0 < \beta \leq 1$, initial

change ΔG_0, and residual ΔG_∞ [1]. The eight-to-thirteen-point decay curves often under-constrain β and τ jointly. The fitted τ is therefore reported only where the fit is well identified. In all cases, the primary timescale is the robust, model-free half-enhancement time $t_{1/2}$, the delay at which the conductance enhancement has fallen to half its initial value. Where both measures are available, they agree. Both are *apparent* timescales determined by the material and by the amplitude used to write and read the state. The composition series and chemistry comparisons each use a fixed protocol because section 4.6 shows that the timescale is protocol-dependent. Fitted β values lie predominantly in the range 0.6–0.9, consistent with stretched relaxation and the heterogeneous distribution of activation barriers invoked in chapter 2.

Data-quality control and aggregation. Two screening layers are applied before any number is reported: erratic or non-physical measurements are flagged and excluded (including the archive's `Y/N` broken flags), and visual review stores curve verdicts (used as measured, cleaned, or discarded) under the full device–measurement–day–pixel key in a curation registry applied automatically by the analysis scripts. All timescales are recomputed after this screening because the instrument-side fit is not robust to anomalous points. The fabricated substrate is the experimental unit: surviving curves are reduced to junction medians, junctions to a substrate median, and substrates to a composition-cell median before inference. A substrate with two measured junctions therefore contributes once. The representative curves of fig. 4.2 are drawn by script from substrates near their cell medians, not chosen as inferential replicates. The raw-to-summary trail, curation precedence, and stretched-exponential identification thresholds are documented in Appendix C.

4.4 Composition Control of the Dynamics

The *Super Yellow*/PEO/LiOTf composition grid varies r_{PEO} over 0.3, 0.6 and 1.2 and r_{salt} over 0.045, 0.09 and 0.18. The pulse and decay datasets contain two to four nominally identical silver-electrode substrates per cell; the HYST dataset contains further compatible batches. Each measurement was acquired under its own fixed protocol (table 4.1), so every composition contrast is drawn *within* one measurement; the 3 V potentiation and 4 V depotentiation data are never pooled. Representative curves in fig. 4.2 show the measurements before they are reduced to on–off ratio, growth exponent, turnover, and fading-memory time. Across the grid, the switching window, potentiation response, and fading-memory timescale change together.

The steady-state switching window contracts as r_{PEO} rises. At fixed $r_{\text{salt}} = 0.09$, the median on–off ratio falls from ≈ 3.3 at $r_{PEO} = 0.3$ to ≈ 2.4 at $r_{PEO} = 1.2$, while the normalised loop area falls from ≈ 0.33 to ≈ 0.18. The activation voltage remains within 2.2 V–2.4 V across the grid. Composition thus changes how *much* the conductance moves, but not the voltage at which it begins to move. The widest window occurs in the low-PEO row, with the largest loop area at the low-salt corner ($r_{PEO} = 0.3$, $r_{\text{salt}} = 0.045$: area ≈ 0.44). The per-sweep on–off ratio is strongly right-skewed there: the cell median is ≈ 3.4, but individual sweeps reach ~ 40. Cell medians are therefore used throughout to prevent these few outlying sweeps from dominating the comparison.

Potentiation also weakens and changes shape with composition (fig. 4.5). The PEO and salt mass ratios affect different features of the response. The per-decade growth of the conductance ratio (the log–log slope α) falls with PEO content, most cleanly at the lowest salt ratio. It drops from $\alpha \approx 1.1$ (near-linear accumulation) at $r_{PEO} = 0.3$ to $\alpha \approx 0.3$ (strongly compressive) at $r_{PEO} = 1.2$. The usable dynamic range follows the same

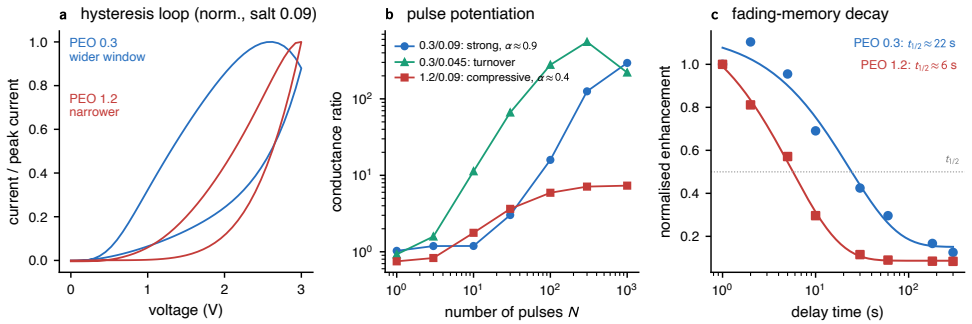


Figure 4.2: Representative raw curves of the three common measurements, on devices chosen near their cell medians (*Super Yellow*/PEO/LiOTf/Ag). (a) Hysteresis loops, each normalised to its own peak current so the comparison is the loop *openness* (the switching window), not absolute conductance: the low-PEO device (PEO 0.3, blue) opens a wider window than the high-PEO one (PEO 1.2, red), which carries more current but switches over a narrower range. (b) Pulse potentiation (conductance ratio vs pulse number, log–log): a strong near-linear accumulation ($\alpha \approx 0.9$), a low-salt cell that *turns over* and then declines, and a weak, compressive high-PEO response ($\alpha \approx 0.4$). (c) Fading-memory decay (enhancement vs delay, normalised to the first read) with stretched-exponential fits and the half-enhancement level dotted: the low-PEO device crosses $t_{1/2}$ far later ($\approx 22\,\mathrm{s}$) than the high-PEO one ($\approx 6\,\mathrm{s}$).

direction: the peak conductance ratio is largest in the low-PEO, low-salt corner ($\approx 480\times$ at $r_{PEO} = 0.3$, $r_{\mathrm{salt}} = 0.045$) and falls to a few-fold at high PEO. Salt content instead governs how long the response grows before saturation. At the lowest salt ratio (0.045), every cell *turns over* by $N \sim 100$–300 pulses; at the highest salt ratio (0.18), the conductance ratio continues to grow through the largest count tested ($N \approx 1000$), with 0.09 intermediate. The α trend is noisier at the higher salt ratios, and the sparsely sampled $r_{PEO} = 0.3$, $r_{\mathrm{salt}} = 0.18$ cell ($n = 2$, curation-recovered) departs from it. Turnover also places a practical ceiling on the usable pulse count. These composition-dependent, predominantly compressive responses provide the nonlinear input encoding used in Chapter 5; the low-salt cells additionally become non-monotone at high pulse counts.

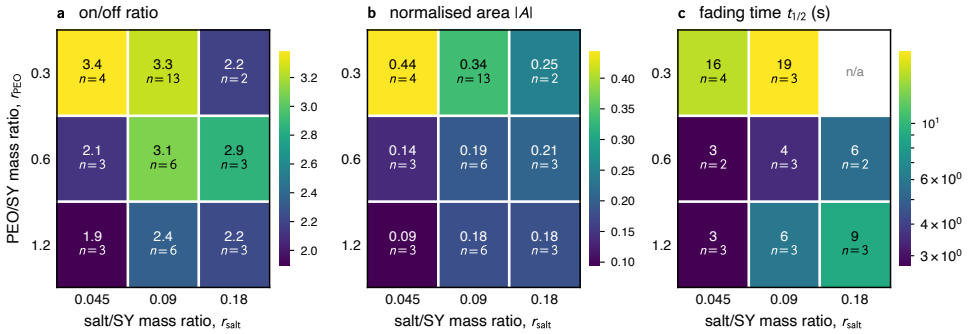


Figure 4.3: Composition control of the steady-state and dynamical response of *Super Yellow*/PEO/LiOTf devices. Across the replicated grid, increasing the PEO-to-*Super Yellow* mass ratio is associated overall with smaller switching windows (a, b) and shorter fading-memory times (c). Individual fixed-salt rows are not strictly monotone, especially in the sparsely sampled salt-rich cells.

The fading-memory timescale varies with composition over roughly 2 s–20 s. The longest retention occurs in the low-PEO row (at $r_{PEO} = 0.3$, $t_{1/2} \approx 19$ s at $r_{\text{salt}} = 0.09$, with an identified $\tau \approx 25$ s); increasing r_{PEO} to 0.6 or 1.2 shortens the timescale to a few seconds. A single curation-recovered substrate at the still-lower $r_{PEO} = 0.15$ level ($t_{1/2} \approx 22$ s) is consistent with this trend, though it lies outside the replicated grid and is not counted toward it. The independently computed instrument-side single-exponential time constant reproduces the same ordering. Within composition cells, the robust model-free timescale has $\sigma(\ln t_{1/2}) \approx 0.26$; the fit-noisier single-exponential τ has a larger spread of about 0.85 (sections C.13 and D.4). The available fabrication covariates do not account for the composition association, but secondary contributions and unrecorded batch effects remain possible (Appendix C). Chapter 5 treats each cell median as a component value and the within-cell spread as its tolerance; the range across the whole device library is mapped in Appendix C. The same composition axis appears in potentiation strength: the growth exponent α falls across the replicated grid, with within-cell

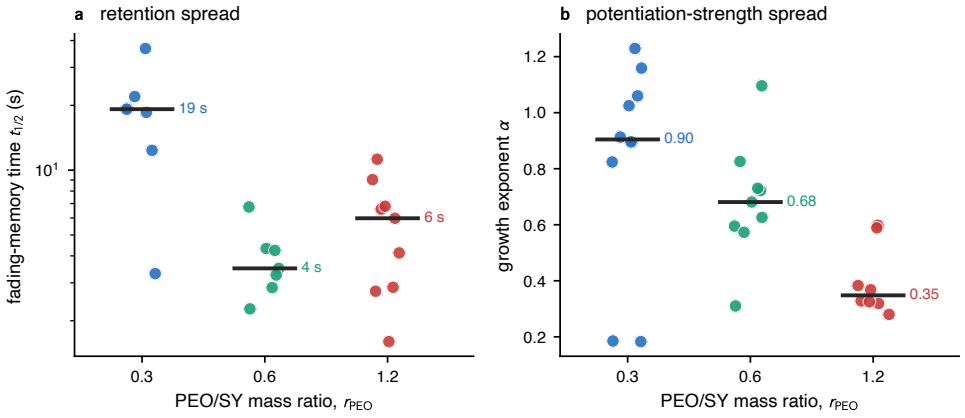


Figure 4.4: Per-substrate distributions underlying the cell medians of fig. 4.3 (*Super Yellow*/PEO/LiOTf/Ag; black bar is the cell median). (a) Fading-memory time $t_{1/2}$ and (b) potentiation growth exponent α, each against r_{PEO}. The PEO-to-SY mass ratio sets the broad level, while within-cell substrate-to-substrate variation is comparable to the steps between cells. Chapter 5 evaluates both sources of variation.

variation around each median.

Across the replicated grid, increasing r_{PEO} is associated with shorter fading-memory time and smaller switching and potentiation responses, while r_{salt} strongly separates turnover. At the substrate level, the four reported gradients are $t_{1/2}$: $\rho = -0.48$, Holm-adjusted $p = 0.022$; growth exponent: $\rho = -0.55$, $p = 0.011$; peak ratio: $\rho = -0.54$, $p = 0.011$; and on–off ratio: $\rho = -0.53$, $p = 0.003$. Permutations restricted within acquisition quarter give the same conclusion (Appendix C). The axes are not fully orthogonal: a batch-adjusted factorial analysis finds no PEO×salt interaction for $t_{1/2}$ at the 5 % level, but does find interactions for growth exponent and peak ratio, so the potentiation effect of r_{PEO} depends on r_{salt}. Bootstrap intervals are reported only for cells with at least three substrates; for $n = 2$, the observed range is reported instead. This replicated composition result is the quantitative core of the chapter; the per-cell values are tabulated in Appendix C. If a more ion-rich

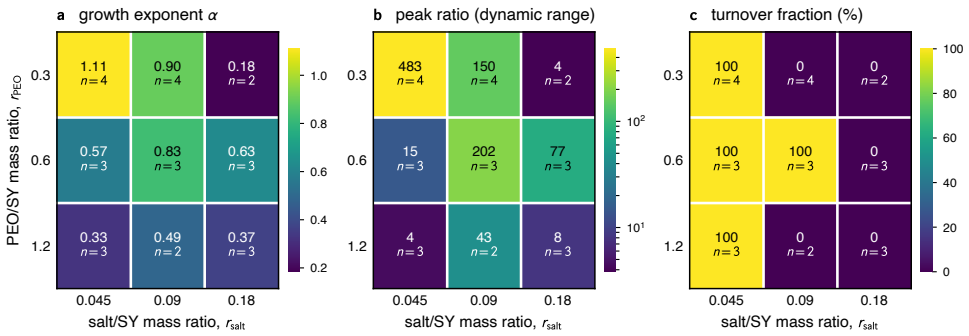


Figure 4.5: Potentiation (pulse-integration) response across the *Super Yellow*/PEO/LiOTf composition grid: (a) the log–log growth exponent α, (b) the peak conductance ratio (usable dynamic range, log scale), and (c) the fraction of substrates whose response turns over within the measured range ($N \leq 1000$). Across the replicated grid, increasing r_{PEO} is associated overall with lower α and dynamic range, although salt-dependent interactions produce non-monotone cells. Increasing r_{salt} suppresses turnover, sustaining growth to higher pulse counts. Per-cell median, with the substrate count n annotated in each cell.

matrix relaxes a displaced ionic profile more readily, its equilibrium ionic resistance should also fall. Section 4.7 tests that prediction by impedance spectroscopy.

4.5 Chemical Tuning of the Dynamics

Beyond composition, the identity of the electrolyte components shifts the dynamics further. Each matched cell in this survey (fixed composition, electrode, and protocol) contains at most one or two screened devices. The magnitudes are therefore indicative, and the sample size is given in every case. Structural and optical probes provide a separate view of the chemistry. Blend films and scratched powders at the lead composition (0.3/0.09) were examined by ATR-FTIR, X-ray diffraction, and UV-Vis absorption to probe ion association, host backbone order, crystallinity,

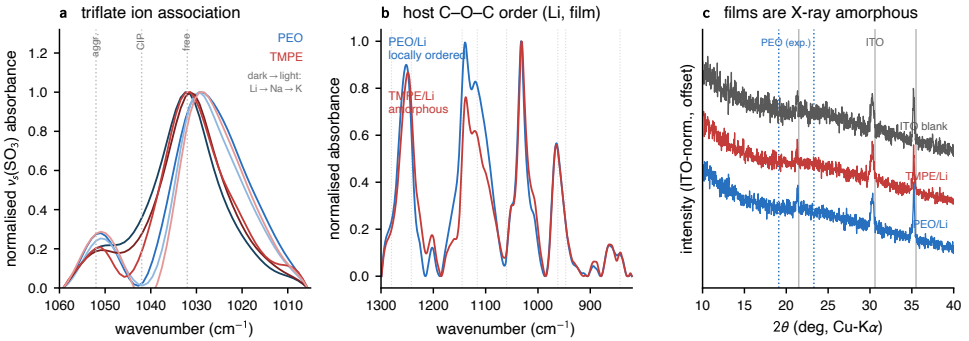


Figure 4.6: Microscopic corroboration of the chemistry axis (electrode-free *Super Yellow*-blend films/powders, 0.3/0.09, one specimen per chemistry). (a) ATR-FTIR triflate ion-association band $\nu_s(SO_3)$, whose position reports the ion-association state (spectroscopically free near 1032 cm^{-1}, contact ion pair near 1042 cm^{-1}, aggregate above; reference positions dotted [2]; PEO hosts in blues, TMPE hosts in reds, darker to lighter for Li/Na/K): it lies at 1029 cm^{-1}–1032 cm^{-1} in every host×cation sample, with no cation-resolved shift. (b) ATR-FTIR C–O–C envelope: the linear PEO host (blue) shows a resolved multiplet, indicating retained *local* (conformational) ether-backbone order, whereas the hyperbranched TMPE host (red) shows a single broad, amorphous band [2]. (c) X-ray diffraction of PEO/Li and TMPE/Li with a bare-ITO blank: all patterns show the crystalline ITO substrate reflections (grey lines; 21.5°, 30.6° and 35.5°), with no sharp PEO crystalline doublet (blue dotted; 19.1° and 23.3°) or crystalline salt phase. The composite films are therefore X-ray amorphous, and the PEO order in (b) is short-range [2, 3]. ATR spectra are normalised after a local/rubber-band baseline and XRD is normalised to the ITO peak; intensities are not comparable across specimens because sampling and thickness differ, so only band positions and shapes are interpreted.

and the semiconductor's electronic gap without driving a device (figs. 4.6 and 4.8). These measurements also use one specimen per chemistry.

Host. Substituting the linear polyether PEO with the hyperbranched polyether TMPE, at fixed lithium-triflate chemistry and composition, shortens the fading-memory time by roughly a factor of six (PEO/LiOTf: $t_{1/2} \approx 20$ s–25 s, three screened silver devices; TMPE/LiOTf: ≈ 4 s, one

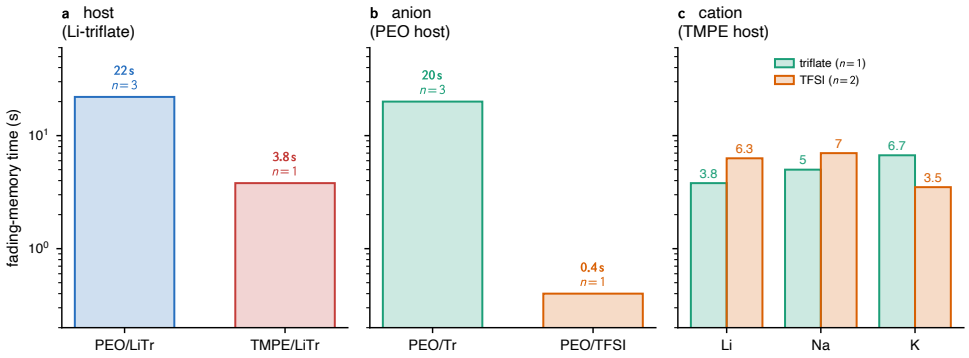


Figure 4.7: Illustrative effect of electrolyte chemistry on the fading-memory time constant (*Super Yellow*, 0.3/0.09, silver, matched 4 V/2 V protocol). Host (PEO vs TMPE, Li-triflate) and anion (triflate vs TFSI, PEO host) produce the largest shifts. The cation panel shows the same three cations in the TMPE host under each anion: potassium is longest in triflate and shortest in TFSI, so the cation produces no consistent, host- and anion-independent ordering. Sample sizes are small (n as labelled); see text.

screened silver device with an identified decay). This comparison has three PEO devices but only one TMPE device. The lithium pair from the designed host×cation batch could not be used because the PEO/LiOTf device was broken. The analysis therefore compares the TMPE device with the nearest-matched silver PEO/LiOTf devices (same operator, fresh semiconductor, glovebox), which differ in fabrication date.

An independent microscopic correlate of the host substitution is visible in the polyether backbone itself, and at two length scales. In ATR-FTIR the linear PEO film shows a resolved C–O–C stretching multiplet, with the weak order-sensitive marker bands near 843 cm^{-1}, 962 cm^{-1} and 1242 cm^{-1}, indicating retained *local* (conformational) order of the ether backbone, whereas the hyperbranched TMPE film collapses these into the single broad band of an amorphous polyether (fig. 4.6b) [2]. At the longer length scale of crystallites, however, X-ray diffraction on the same sample set finds *neither* host crystalline in the composite: every pattern

is dominated by the ITO substrate (confirmed against a bare-substrate blank), with no sharp PEO crystalline reflections and no segregated crystalline salt phase (fig. 4.6c), so the films are X-ray amorphous and PEO retains short-range order without forming long-range crystallites [2, 3]. A host that is locally more ordered, even while globally amorphous, set against a fully amorphous TMPE, is consistent with the lower segmental mobility and slower ionic relaxation that the *longer* PEO fading-memory time would require: in polyether electrolytes ion transport is carried by segmental motion of the host above its glass transition, so the relaxation of a displaced ionic profile is rate-limited by that segmental mobility [4–6]. With one specimen per host this is offered as a structural correlate, not a demonstrated cause. The host ordering is moreover not specific to the silver electrode: on the independent gold-electrode corpus at the same composition, PEO again retains far longer than TMPE and supports a markedly wider switching window and stronger potentiation (section 4.8), so the host effect survives a change of electrode. That the salt forms no crystalline phase in any chemistry additionally supports reading these blends as homogeneous, salt-dissolved ion-conducting media.

Anion. Replacing triflate with the more charge-delocalised TFSI anion in the PEO host reduces the fading-memory time from $\approx 20\,\mathrm{s}$ to approximately 1 s. This direction is consistent with the electrolyte literature: charge delocalisation in TFSI weakens ion pairing and lowers salt lattice energy, promoting dissociation and plasticisation of the polyether host and increasing ionic mobility [7, 8]. The TFSI decays also tend toward a compressed shape, whereas the triflate decays are stretched, suggesting different relaxation-time distributions. This comparison is weakly controlled: the single PEO/TFSI device belongs to a later fabrication generation and also differs in spin speed and salt filtration. The approximately twenty-fold difference is therefore reported as illustrative.

Cation. The data do not support the hypothesised $Li^+ > Na^+ > K^+$ ordering of fading-memory time derived from the hard–soft acid–base argument of chapter 2 [9]. Two cation series use the same TMPE host, composition, electrode, and protocol, and differ in the salt anion (fig. 4.7). With triflate, the timescale increases from Li^+ ($\approx 3.8\,s$) to Na^+ ($\approx 5.0\,s$) and K^+ ($\approx 6.7\,s$); each value comes from one screened device with $R^2 \geq 0.93$. With TFSI, potassium is shortest ($\approx 3.5\,s$), while lithium and sodium are longer ($\approx 6\,s$–$7\,s$, two devices per cation). Thus, cation identity shifts the timescale by roughly a factor of two within a fixed anion, but the ordering changes with the anion and within-cation variation is comparable to between-cation differences. These sample sizes cannot establish a host- and anion-independent ordering. A powered test would require a dedicated series at fixed host, anion, protocol, and electrode with at least three replicates per cell (section 4.10).

Independent probes provide additional constraints. The triflate symmetric SO_3 stretch $\nu_s(SO_3)$ lies at $1029\,cm^{-1}$–$1032\,cm^{-1}$ in all six host×cation films and powders, a spread at or below instrumental resolution with no cation ordering (fig. 4.6a). UV-Vis absorption of *Super Yellow*/TMPE films places the *Super Yellow* π–π^* peak [10] at 446 nm for Li^+, Na^+, and K^+ (fig. 4.8). These results indicate no resolved cation effect on ion-pairing state or semiconductor optical gap; neither measurement tests the ordering of retention directly. On gold electrodes, the TMPE/triflate steady-state window is also flat across the three cations (on–off ratio ≈ 1.3–1.5, no monotone trend; section 4.8).

In these illustrative comparisons, host and anion substitutions produce larger shifts in fading-memory time than cation substitution.

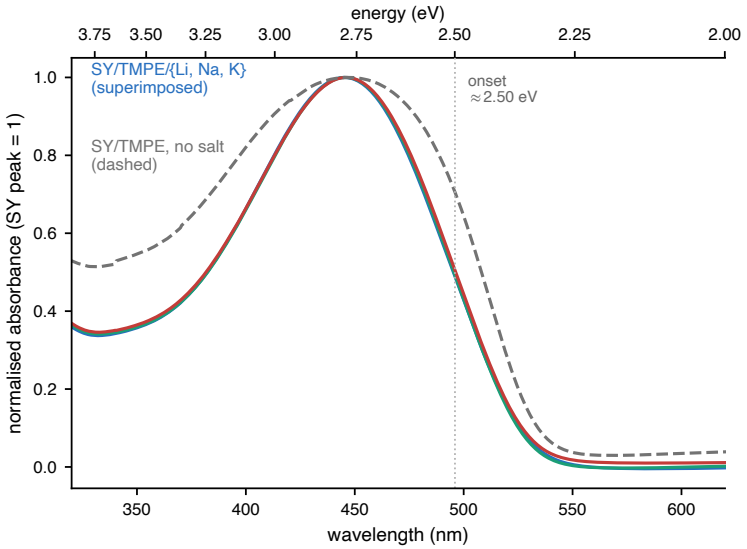


Figure 4.8: UV-Vis absorption of *Super Yellow*/TMPE blend films (lead composition 0.3/0.09; Au-electrode 2025 generation; one specimen per chemistry), normalised to the *Super Yellow* π–π^* peak. The peak (≈ 446 nm) and the absorption onset (≈ 2.50 eV) are superimposed for Li^+, Na^+, and K^+; the salt-free control (dashed) coincides at the band edge (its differing blue side and lower absolute absorbance are thickness/scattering, not electronic). The cation does not shift the semiconductor's optical gap. Only band *positions* are read because absorbance magnitudes are thickness-confounded.

4.6 Protocol-Dependent Timescale

The drive protocol constrains every comparison above. The read voltage exceeds the ionic activation threshold, and write amplitude controls the displaced ionic profile; both can change the apparent fading-memory time. In one device measured under both protocols, the fitted relaxation time increases from ≈ 4.6 s under a 3 V write and 1.5 V read to ≈ 15.5 s under a 6 V write and 3 V read (fig. 4.9).

One interpretation is that a larger write amplitude displaces more ionic charge and, at the largest amplitudes, may introduce electrode-level

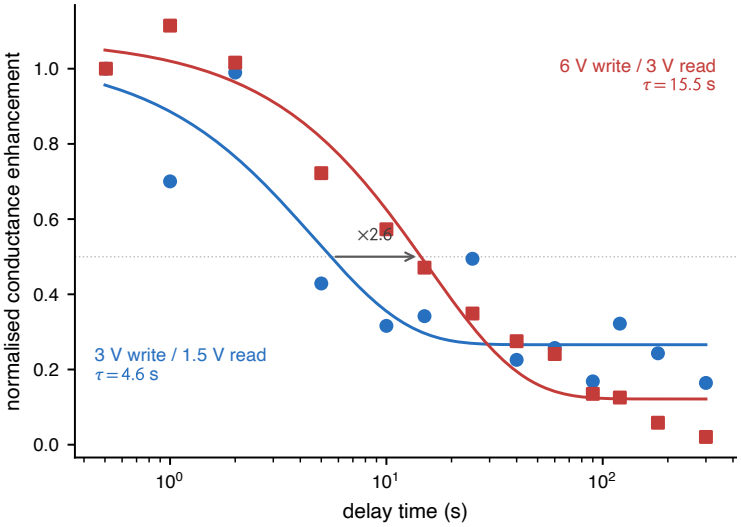


Figure 4.9: The apparent fading-memory time is set in part by the drive protocol. The same device, measured under a larger write/read amplitude, shows a markedly longer apparent relaxation time: the fitted single-exponential time constant rises from $\tau \approx 4.6\,\mathrm{s}$ to $\approx 15.5\,\mathrm{s}$, and the half-enhancement crossing (dotted, arrow) shifts by $\approx 2.8\times$.

electrochemical processes, producing a longer-lived written state. The supra-threshold read may also perturb the state during interrogation. The composition result of section 4.4 remains internally comparable because it uses one fixed protocol. Chemistry comparisons that span device generations are more exposed to protocol differences. Material and drive conditions therefore co-determine the apparent volatile timescale; cross-device or cross-chemistry comparisons must match the write/read protocol, electrode, and composition.

The same lever reappears, independently, on the gold-electrode corpus of section 4.8. There, matched-amplitude negative pulse trains sweep the pulse count and the inter-pulse interval together, with the relaxation recorded in the same run, and both halves of the protocol dependence are observed directly: potentiation grows steeply as the inter-pulse interval

shortens (rank correlation $\rho \approx -0.94$ to -0.99, $p < 0.001$, in three independent series of 9–11 matched runs), and the depth of the written state co-sets the subsequent relaxation time ($\rho \approx +0.46$ between potentiation ratio and decay half-time, $p = 0.015$, $n = 28$ runs). The drive–retention coupling that this section infers from a two-protocol contrast is therefore seen *within single experiments* on a second electrode (Appendix C).

4.7 Microscopic Corroboration: Impedance Spectroscopy

Every result so far rests on the *driven* response (pulses and decays). Small-signal impedance spectroscopy offers an independent test of the mechanism proposed for the composition trend: if increasing r_{PEO} shortens the fading-memory time because a more ion-rich matrix lets the displaced ionic profile relax more readily, then the *equilibrium* ionic resistance should fall along the same composition axis. The archive contains such a measurement, acquired on the same substrates: 0 V-DC spectra (40 mV AC, 1 Hz–1×10^6 Hz) on 26 silver-electrode *Super Yellow*/PEO/LiOTf substrates spanning the composition grid.

Each spectrum is summarised in two separate ways. The first is *model-free*: the real part of the impedance at the apex of the main Nyquist semicircle, $Z_{\mathrm{real}}^{\mathrm{apex}}$, which fixes the resistance scale of the spectrum without committing to any circuit. The second is *model-based*: the bulk ionic resistance R_{ion} of an equivalent circuit adapted from the polymer light-emitting electrochemical cell analysis of Munar *et al.* [11], reported only as corroboration of the model-free quantity because the per-spectrum circuit fits are genuinely ill-conditioned. The circuit topology, its two automated fits, and the per-cell values of both observables are detailed in Appendix C.

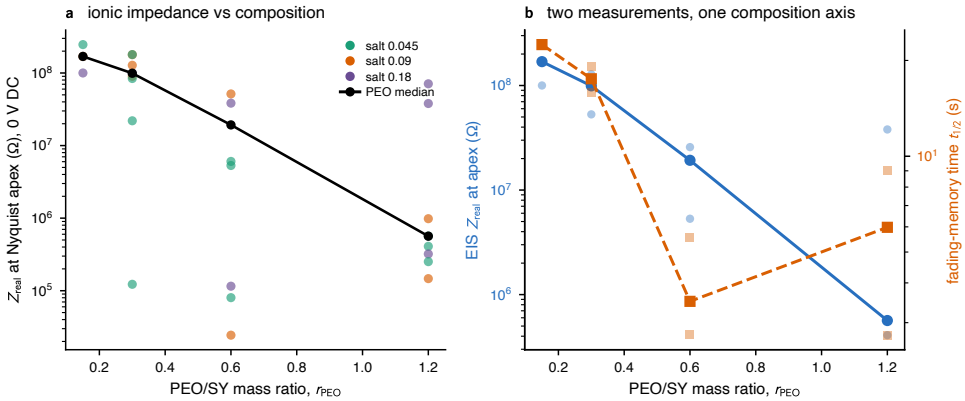


Figure 4.10: Equilibrium (0 V-DC) impedance spectroscopy corroborates the composition control of the dynamics (silver *Super Yellow*/PEO/LiOTf grid). (a) The impedance resistance scale (here the model-free real impedance at the Nyquist apex) falls by more than three orders of magnitude as r_{PEO} rises (points are substrates, coloured by r_{salt}; the black line is the r_{PEO} median). (b) The equilibrium impedance scale (left axis) and the *independently* measured fading-memory time $t_{1/2}$ (right axis) decline along the *same* PEO-to-SY composition axis (cell-level points with per-r_{PEO} median trends): the two are co-driven by composition, not independent variables.

Both observables fall steeply with r_{PEO} (fig. 4.10a). The model-free $Z_{\text{real}}^{\text{apex}}$ drops by more than three orders of magnitude across the measured series (cell medians from $\sim 10^8\,\Omega$ at $r_{PEO} = 0.15$–0.3 to $\sim 10^5\,\Omega$ at $r_{PEO} = 1.2$), with a rank correlation against r_{PEO} of -0.64 per substrate ($n = 26$) and -0.83 on cell medians; the fitted R_{ion} gives the same picture (-0.84 on cell medians). This is the direction the mechanism requires: more ion-transport polymer relative to *Super Yellow*, lower ionic resistance.

The EIS resistance scale and the *independently* measured fading-memory time fall along the *same* composition axis (fig. 4.10b): both decline together as r_{PEO} rises, and across cells $\log Z_{\text{real}}^{\text{apex}}$ covaries with $\log t_{1/2}$ at $r \approx +0.73$ ($+0.60$ per substrate, $n = 23$). Three qualifications bound this reading:

- The PEO-to-SY mass ratio is a shared covariate of both quantities. Their correlation therefore shows that equilibrium impedance and driven retention move together along the composition axis; it is not an independent causal test.

- The resistance *scale* carries the composition signal, whereas the EIS relaxation *frequency* does not: the apex frequency corresponds to a sub-second RC time and does *not* track the seconds-long fading-memory time. This separates the small-signal ionic-transport bottleneck reported by impedance spectroscopy from the slower large-signal relaxation that determines the fading-memory response. As an order-of-magnitude check, scaling the median apex time ($\approx 0.089\,\mathrm{s}$) by the low-frequency-to-geometric capacitance ratio (≈ 69) gives $\approx 6.2\,\mathrm{s}$, close to the median $t_{1/2}$; this consistency does not establish that the two measurements share one physical clock (Appendix C).

- $Z_{\mathrm{real}}^{\mathrm{apex}}$ spans $\sim 10^4\times$ while film thickness spans only $2.6\times$, and the PEO–impedance association persists after adjustment for thickness (partial correlation -0.60). This makes thickness alone insufficient to explain the impedance gradient, although it does not exclude secondary geometric effects (section 4.2).

Impedance spectroscopy thus provides an independently measured equilibrium quantity consistent with the ionic-resistance part of the mechanistic picture below; it does not uniquely identify that mechanism. The same spectra bear on where the seconds-scale storage sits. Their low-frequency capacitance exceeds the geometric capacitance by a median factor of about 69 (Appendix C). This is consistent with the relaxation time arising from ionic storage within the $0.0825\,\mathrm{cm}^2$ film; no external capacitor is involved, although the low-frequency estimate is model-dependent.

4.8 Electrode and Polarity Dependence

Every result so far uses the silver top electrode. The archive also contains 50 gold-electrode devices, treated here as a third illustrative axis. Every gold device uses the single lead composition (PEO or TMPE host, 0.3/0.09); this corpus therefore contains no information about the composition series of section 4.4, which remains a silver-only result. It instead contrasts electrochemically active silver with inert gold at matched chemistry and provides an independent-electrode check on the chemistry comparisons of section 4.5.

At matched chemistry and composition, and under the canonical positive-polarity protocols used everywhere above, the electrode shifts all three behavioural families, and does so coherently (fig. 4.11). In the lead PEO/LiOTf cell, replacing silver by gold narrows the steady-state switching window (median on–off ratio $3.3 \rightarrow 2.0$, normalised loop area $0.33 \rightarrow 0.12$), weakens and flattens potentiation (peak conductance ratio $\approx 124\times \rightarrow 29\times$, log–log slope $\alpha \approx 0.9 \rightarrow 0.45$), and raises the activation voltage ($\approx 2.4\,\mathrm{V}$–$2.7\,\mathrm{V}$), yet *lengthens* the fading-memory time, from $t_{1/2} \approx 19\,\mathrm{s}$ on silver (three devices) to $\approx 87\,\mathrm{s}$ on the two clean gold devices ($R^2 \approx 0.97$–0.99, $\beta \approx 0.6$). The same window-and-potentiation contraction holds in the TMPE/LiOTf cell (on–off $3.8 \rightarrow 1.3$; peak $54\times \rightarrow 1.9\times$), again at positive drive. The per-cell metrics for every chemistry on both electrodes, including the host-free and salt-free negative controls, are tabulated in Appendix C.

This difference is consistent with the contrast between active and inert electrodes. Silver can inject and electrodeposit mobile Ag^+, the basis of electrochemical-metallisation switching, and may therefore add a large volatile component: a wider window, steeper potentiation, lower threshold, and faster relaxation [12, 13]. Gold is noble and inert, leaving the polymer-electrolyte ionic mechanism to produce a smaller, higher-threshold, and

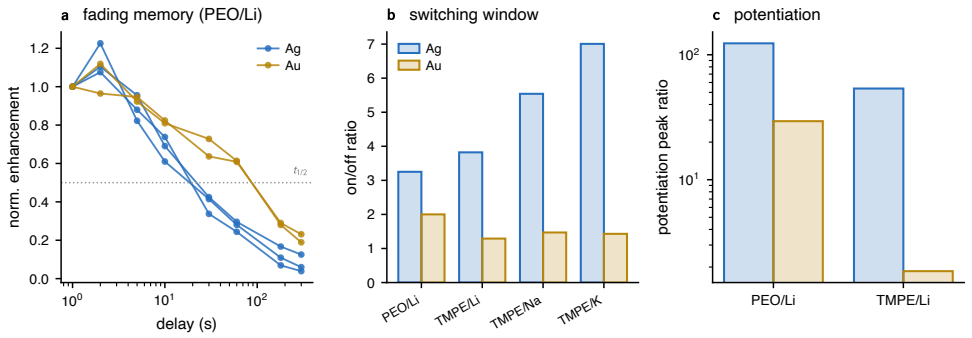


Figure 4.11: Illustrative electrode contrast at the lead composition (0.3/0.09), under the canonical positive-polarity protocols (for the negative-polarity behaviour of the gold devices see fig. 4.12). (a) Fading-memory decays of PEO/LiOTf devices, normalised to the first read: the gold curves (gold) cross half-enhancement later than the silver ones (blue), i.e. gold retains *longer*. (b) The steady-state switching window (on–off ratio) is narrower on gold for every chemistry, and is flat across the TMPE cation series on gold (no Li/Na/K ordering). (c) Pulse-potentiation dynamic range (peak ratio, log scale) is far smaller on gold. The inert electrode thus gives a weaker, higher-threshold, but more retentive response. Small n per cell, no human curve curation on the gold side, and a generation offset between the corpora bound the comparison (see text).

more persistent response. The higher activation voltage on thinner gold devices supports an interfacial electrode contribution over a film-thickness explanation [14].

Negative-polarity response of the gold devices. Additional campaigns on the same Au/TMPE/triflate generation measured negative bias. These data were recovered from outside the curated archive and analysed with a dedicated parser; their sources, validation against catalogued sweeps, and per-file tables are documented in Appendix C. The devices are strongly polarity-asymmetric: within a junction, bipolar sweeps carry ≈ 2.5–$42\times$ more current at -3 V than at $+3$ V (fig. 4.12a), with the same direction across the catalogued negative-going sweeps. Under negative drive, the gold devices show a rate-dependent switching window. Across

sweep rates of $\sim$0.003–0.4 V s^{-1}, the on–off ratio rises as the sweep slows (rank correlations down to -0.99, $p < 0.001$), from $\approx$1 at the standard rate to 20–31 in the slowest -3 V sweeps (fig. 4.12b). Constant-bias holds locate an activation threshold between -1.5 and -2.5 V: current decays below this range and grows by one to three orders of magnitude above it over tens to hundreds of seconds. Negative pulse trains potentiate by one to two orders of magnitude at a matched protocol (-5 V pulses, -2.25 V read: cation medians $\approx$ 40–230$\times$), while positive pulses depress the written state. The response is therefore bidirectionally programmable under this negative-polarity protocol.

These campaigns add fading-memory measurements for the TMPE host on gold. Sparse delayed reads after the pulse trains, with the device floating between reads, follow strongly stretched exponentials ($\beta \approx 0.33$–0.5) with model-free half-decay times of ≈ 0.4 s–21 s across the three cations (fig. 4.12c). These values are on the same seconds scale as TMPE on silver and below the ≈ 87 s measured for PEO on gold, providing a gold-electrode comparison of host retention. Under the matched negative protocol, all three cations have overlapping potentiation ranges and relaxation times, with substrate-to-substrate spread comparable to any cation difference. The concentration of the response in one bias direction after removal of the active Ag^+ pathway is consistent with an interfacial role for the dissimilar ITO and gold contacts and their accumulated ion profiles, but the electrical data do not establish that mechanism.

The second electrode reproduces two observations from section 4.5: PEO retains longer than TMPE, and the switching window has no transferable cation ordering. The gold corpus also provides negative controls absent from the silver set. Host-free (semiconductor-plus-salt) and salt-free (semiconductor-plus-host) films show neither a switching window (on–off ≈ 1.0) nor potentiation (peak ≈ 1), indicating that both the ion-transport polymer and dissolved salt are required for the memristive response.

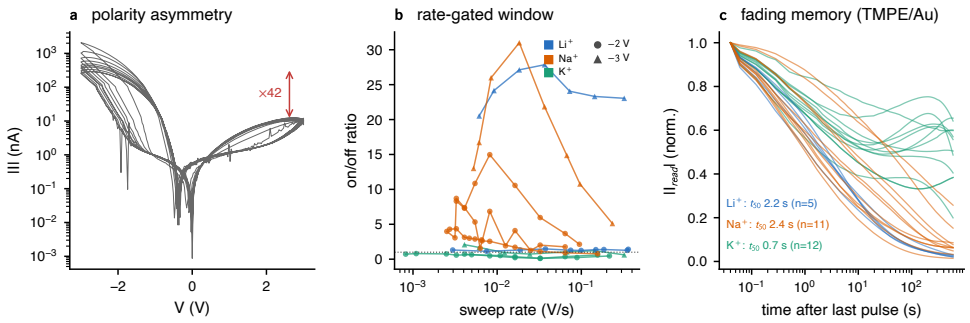


Figure 4.12: Negative-polarity response of the gold/TMPE/triflate devices (illustrative tier; Appendix C). (a) Exemplar bipolar sweep (±3 V, ten cycles, Na^+ device): the negative branch carries ≈ 42× the current of the positive one. (b) On–off ratio against sweep rate for all step-count series (colour = cation, marker = amplitude): slow sweeps give ratios of 20–31, while the standard fast protocol gives ≈1. (c) Sparse-read relaxation after −5 V pulse trains, normalised to the first read, with per-cation median half-decay times. The times are on the seconds scale, as for TMPE on silver, and below those of PEO on gold.

Frequency-domain response: a bias-programmed low-frequency amplifier. A third corpus measures the same Au/TMPE/triflate chemistry in the frequency domain: 549 lock-in sweeps from 1 Hz to 500 kHz on a fresh batch, with one substrate per cation tracked from day 1 to day 23. The junction is placed in series with a 4 nF capacitor and driven by a small sinusoid superimposed on a programmable DC offset. The apparatus, channel decode, and per-file tables are documented in Appendix C. Above ∼10 Hz, the unbiased junction behaves as a passive linear dielectric: the transfer ratio follows the capacitive divider and is amplitude-independent to within a median 0.2 % from 0.5 to 2.4 Vpk. This boundary is consistent with the rate dependence in fig. 4.12b. In the same plateau, junction capacitance increases Li < Na < K by ≈1.4× across the same-batch trio, compared with an ≈1 % pixel spread. Same-recipe profilometry makes a thickness origin less likely, but one substrate per salt limits this result to a candidate dielectric cation effect; profilometry on matched specimens

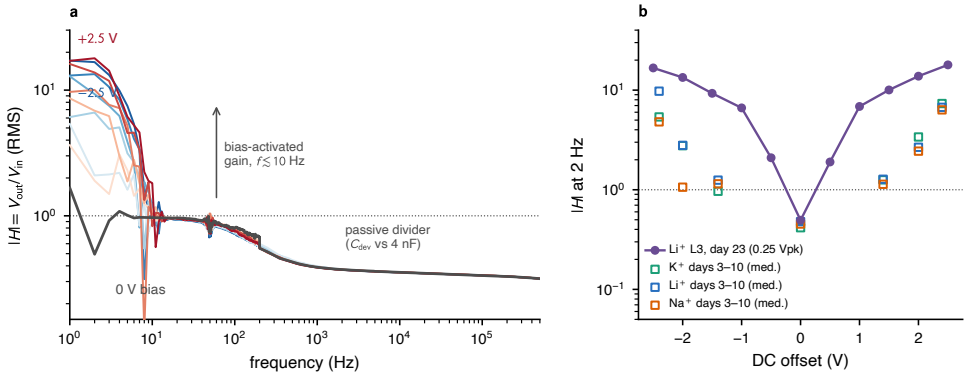


Figure 4.13: Frequency-domain response of the gold/TMPE/triflate devices (illustrative tier; Appendix C). (a) Transfer ratio $|H|(f)$ of one Li^+ junction in series with 4 nF (day 23, 0.25 Vpk) at eleven DC offsets from 0 (grey) to ±2.5 V (blue negative, red positive). Above ∼1 kHz, all curves follow the passive capacitive divider; below ∼10 Hz, bias produces a fundamental-frequency gain of up to ≈18, increasing with bias magnitude and nearly symmetric in sign. (b) $|H|$ at 2 Hz against DC offset for the day-23 series (line) and per-cation medians of earlier sessions (squares; days 3–10, 0.4 Vpk–1.4 Vpk). Gain increases with bias and occurs for all three cations.

would test it directly. Under DC bias, the fundamental-frequency transfer below ∼10 Hz increases from $|H| \approx 0.5$ without bias to ≈18 at ±2.5 V (fig. 4.13). The bias-modulated ionic conductance transfers energy from the DC offset into the signal band. This gain occurs for all three cations at every age tested, is confined to the sub-10 Hz band, and is nearly symmetric in bias polarity (Appendix C). A biased junction therefore operates as a voltage-programmable low-pass amplifier over a band that overlaps the physiological signals analysed in Chapter 5.

All three gold corpora share the same illustrative-tier bounds, which the SI states for each corpus (Appendix C). Each uses the single lead composition on one or two substrates per cation; the curves are uncurated; and the supra-threshold negative read makes decay times protocol-specific, allowing only order-of-magnitude comparison with silver. The 2024–2025

gold generation is also partly confounded with electrode because the silver chemistry was fabricated earlier and the co-fabricated silver batch was not measured under the decay protocol. Electrode and polarity are therefore treated as illustrative effects and cross-electrode checks on the chemistry axis. A matched-generation batch split between silver and gold is required for a quantitative electrode comparison.

4.9 Discussion

Composition is the only replicated axis for control of the volatile dynamics. Increasing the ion-transport-polymer-to-semiconductor mass ratio shortens the fading-memory time and is accompanied by smaller switching and potentiation responses. Host and anion substitutions shift the timescale over a wider range, but those comparisons are illustrative. Cation identity also shifts the dynamics, although the direction changes with host, anion, and protocol at the available sample sizes; the predicted $Li^+ > Na^+ > K^+$ order is therefore not transferable across the dataset.

The electrode acts jointly with drive polarity. Relative to active silver, inert gold weakens switching and potentiation and lengthens retention under the canonical positive drive. Under negative drive, the same gold devices show a full, rate-gated, bidirectionally programmable response (section 4.8). The frequency measurements locate the corresponding spectral boundary: above $\sim 10\,\mathrm{Hz}$ the gold junctions behave as passive linear capacitors at every tested amplitude, whereas a DC bias produces programmable low-frequency gain below that boundary.

The hard–soft acid–base model of chapter 2 remains a qualitative interpretation. It anticipates the cation ordering in the TFSI family, where the most weakly coordinating cation relaxes fastest, but the triflate family runs in the opposite direction. The present data therefore do not support

a quantitative cation law.

Mechanistic interpretation. The mechanism remains a hypothesis. Increasing r_{PEO} does not produce the decrease in absolute on-state conductance expected from simple dilution of the *Super Yellow* percolation pathway; across the replicated grid the association is positive ($\rho = +0.61$, $p < 0.001$, $n = 43$ substrates; Appendix C). The cell-median conductance is not strictly monotone from $r_{PEO} = 0.6$ to 1.2, so this rejects a simple dilution picture rather than establishing a unique transport law. A raised conductance floor is one interpretation consistent with the contracting window: the ion-rich phase may add a parallel, weakly switchable path, with off-state conductance rising faster than on-state conductance. Faster relaxation of the displaced ionic profile could account for the accompanying decrease in fading-memory time. Salt content may set the number of mobile, chargeable states available before saturation, consistent with the later turnover at higher r_{salt}.

Host and anion effects may arise from cation–ether coordination, ion pairing, and segmental mobility [5, 8], which set the distribution of local activation barriers behind stretched-exponential relaxation [1]. Equilibrium impedance supports the ionic-resistance part of this interpretation: it falls along the composition axis and tracks the fading-memory time (section 4.7). ATR associates the host effect with local ether-backbone order, retained in PEO and absent in amorphous TMPE, while XRD shows that every composite remains X-ray amorphous and that the salt is dissolved rather than segregated. Across the cation series, ATR shows no corresponding ion-association fingerprint and UV-Vis shows no shift of the *Super Yellow* optical gap (section 4.5). These measurements are consistent with electrolyte-controlled ion transport rather than a change in the *Super Yellow* electronic backbone.

The anion and barrier-distribution links have not been established micro-

scopically. They require temperature-dependent relaxation measurements and chemistry-resolved impedance (section 4.10), so the hard–soft acid–base interpretation remains qualitative.

For circuit use, the composition grid is a set of component values. Across compositions and nominally identical substrates, the timescale spans roughly an order of magnitude. A descriptive two-σ separation heuristic on the cell-median span $\Delta \ln t_{1/2} = 1.95$, relative to within-cell scatter, gives approximately two to five composition bands. That is the number of time constants that can currently be selected by formulation and told apart despite substrate-to-substrate scatter; it is not an independently measured information-capacity bound (Appendix C). The same spread could also populate a physical reservoir with distinct temporal responses [15–17]. Each junction reproduces its switching window to $\approx 15\,\%$ sweep-to-sweep (Appendix C), substantially less than the diversity across the bank. Figure 4.14 places every screened silver substrate in a fading-memory ($t_{1/2}$) versus dynamic-range (peak ratio) plane. The PEO-to-SY mass ratio sets the broad region occupied by each substrate, so one composition selects a paired memory/nonlinearity operating point; the remaining within-composition scatter is the tolerance on that operating point. Chapter 5 propagates both into the simulations.

4.10 Limitations and Future Work

Statistical power is limited outside the composition axis. Only the *Super Yellow*/PEO/LiOTf composition grid is replicated; the host, anion, and cation comparisons rest on one or two screened devices per matched cell.

1. The cation hypothesis is not tested here at adequate power. The cation does shift the dynamics, but not as predicted: the triflate retention series runs K>Na>Li, the reverse of the hypothesised

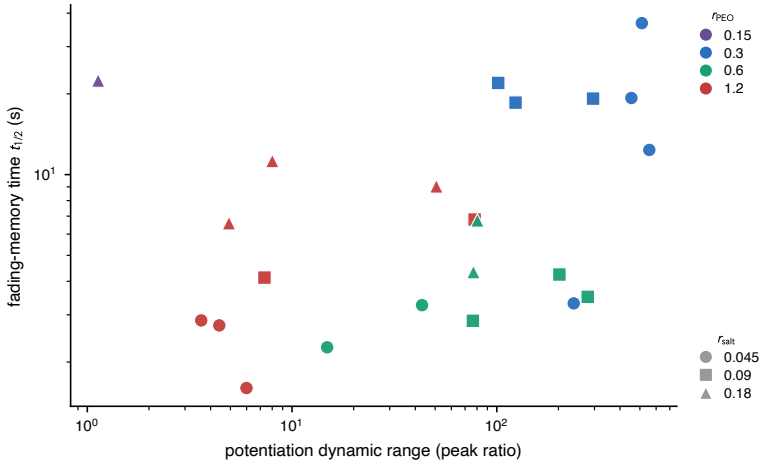


Figure 4.14: Per-substrate design space of the *Super Yellow*/PEO/LiOTf/Ag composition grid: fading-memory time $t_{1/2}$ against potentiation dynamic range (peak ratio), coloured by r_{PEO} and marked by r_{salt} (one point per fabricated substrate, median over its junctions). The PEO ratio selects the broad region a substrate occupies: low PEO (blue) in the long-memory, high-dynamic-range corner, high PEO (red) in the opposite one, so memory and nonlinearity are paired and cannot be set independently. The within-composition scatter is the substrate-to-substrate tolerance that Chapter 5 propagates into its simulations.

$Li^+ > Na^+ > K^+$, and re-orders again with the anion, so the hypothesised ordering is not what the data show and no transferable cation law can be fixed at one or two devices per cell.

2. The supra-threshold read and the amplitude dependence of the apparent timescale (section 4.6) mean that absolute retention values are protocol-specific.

3. The electrode is reported as an illustrative axis (section 4.8), but it remains partly confounded with fabrication generation, and device ageing is an uncontrolled variable that was not modelled.

4. Pulse integration and state relaxation were characterised *separately and at different write amplitudes*: the potentiation curves used a

3 V write and the fading-memory decays a 4 V write (table 4.1), at an effectively common inter-pulse interval. On the lead silver composition, the archive contains no experiment that varies input timing to probe their interaction directly. A matched-amplitude experiment exists on the gold/TMPE corpus (section 4.8), but it uses a different host, electrode, polarity, and read, so it establishes the direction of coupling rather than the lead-cell parameters.

5. The fading-memory protocol measured a single junction per device, so repeat-decay reproducibility of $t_{1/2}$ is not characterised at the population level (only one device carries two junctions; Appendix C). Within-device reproducibility is established only for the steady-state switching window, from its repeated sweeps.

Because drive amplitude affects the apparent timescale (section 4.6), combining the 3 V potentiation shape with the 4 V decay in Chapter 5 requires an explicit modelling assumption that the regimes are compatible. A matched-amplitude experiment that varies both pulse count and inter-pulse interval would test this assumption. Its gold/TMPE negative-polarity counterpart shows coupling between the two measurements (section 4.6); the lead-composition silver experiment remains future work.

A dedicated cation series in one host and anion, at a fixed protocol and electrode, with a sub-threshold read voltage (as in chapter 2) and at least three replicates per cell, would turn the illustrative chemistry survey into a powered comparison. The impedance study of section 4.7 could also be extended beyond the replicated PEO/LiOTf axis at 0 V DC. The archive contains unmodelled bias-resolved spectra up to 3 V DC and chemistry-varied devices; analysing them would test the proposed host, anion, and barrier-distribution effects. A matched-generation batch split between silver and gold top electrodes would isolate the electrode effect

from fabrication generation.

Temperature-dependent relaxation measurements are needed to test the proposed role of segmental mobility and distributed activation barriers. A $\tau(T)$ series yielding an Arrhenius or Vogel–Fulcher–Tammann activation energy would connect the host and anion differences to a segmental-relaxation barrier and test the temperature dependence of the stretch exponent β. Encapsulation and systematic ageing measurements would address the remaining stability questions.

4.11 Inputs to Temporal Computing

In the replicated *Super Yellow*/PEO/LiOTf grid, composition tunes the fading-memory time over roughly 2 s–20 s, with the switching window and potentiation strength moving in the same direction. Host and anion contrasts extend the measured range, although each rests on at most one or two devices per matched cell. Cation identity also shifts the dynamics, but the available data do not establish the predicted $Li^+ > Na^+ > K^+$ order as a transferable law. Drive protocol co-sets the apparent timescale and must remain fixed in quantitative comparisons.

Chapter 5 uses the relaxation kernel of each cell ($t_{1/2}$ and, where identified, β) as a component value and the substrate-to-substrate spread as its tolerance. It also uses the pulse-update curves, including their saturation and turnover. Because r_{PEO} changes both the fading-memory time and the input nonlinearity α (fig. 4.14), each composition defines a paired operating point rather than two independent settings. The salt-to-SY mass ratio provides a partly separate control of dynamic range and turnover. The simulations use these values in two ways: single cells and pairs of cells as designed filters, and banks of one or several compositions as reservoirs.

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